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The FaBRIC Strategy for Verifying Neural Feedback System (CDC-1445)

The paper proposes a verification framework called FABRIC for neural feedback systems (NFS). The approach combines forward and backward reachability analysis using polyhedral enclosures and MILP-based formulations. The authors introduce a domain refinement algorithm (DRIPy) to improve backward reachable set approximations and further propose methods for computing both outer and inner reachable sets. The framework is evaluated on benchmark systems, including TORA, Unicycle, and Attitude control systems, demonstrating improvements in computational efficiency and approximation tightness compared with existing approaches such as HyBReach-MILP and BURNS.

Strengths:

- The integration of forward and backward reachability into a unified verification framework (FABRIC) is novel, meaningful, and relevant for neural feedback system verification.
- The methodology is technically solid, incorporating MILP formulations, polyhedral enclosures, and domain refinement techniques.
- The experimental evaluation on standard benchmark problems is comprehensive and demonstrates improvements in runtime and reachable set approximation quality.
- The proposed approach addresses important limitations in existing methods, particularly the over-conservatism in forward analysis and scalability challenges in backward analysis.

Comments and Concerns:

Comment 1: In Eq. (1), the system dynamics are discretized using an Euler-type update. However, the paper does not discuss how discretization errors affect the reachable set computation, especially when sets are propagated over multiple time steps. Clarifying the impact of numerical discretization on the soundness and tightness of the reachability analysis would improve the correctness and reliability of the proposed framework.

Comment 2: Although the formulation of the must and may backward reachable sets in Eqs. (2)–(3) follows standard definitions, the paper does not establish a formal relationship between these sets and the forward reachability conditions in Eqs. (4)–(5). In particular, it remains unclear whether the FABRIC composition guarantees soundness for the overall reach-avoid verification problem or merely provides a heuristic improvement.

Comment 3: The MILP-based backward reachability formulation in Eqs. (6)-(7) optimizes each state dimension independently, implicitly assuming that tight bounds can be obtained through axis-aligned projections. However, this assumption may introduce significant conservatism in coupled dynamical systems. The paper does not quantify this effect or compare the proposed formulation with joint optimization approaches that explicitly account for state coupling.

Comment 4: Algorithm 1 (DRIPy) introduces a volume-based refinement loop, but the stopping criterion lacks theoretical justification. Specifically, the paper does not provide

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guarantees that the iterative domain restriction converges to a fixed point or avoids pathological cases where refinement stalls without meaningful improvement. Furthermore, the choice of the parameter α and its influence on computational cost and approximation quality are not discussed, making the refinement process difficult to reproduce and tune in practice.

Comment 5: The inner-approximation procedures (CRISP/CLEAN) rely on sampling within polyhedral sets. However, the paper does not provide probabilistic guarantees, such as confidence levels or coverage bounds, regarding the computed safe sets. This raises concerns about the reliability of the obtained inner approximations in safety-critical verification scenarios.

Comment 6: Although the experimental results presented in Tables I-III show improvements in volume reduction and runtime, the performance gains are not uniform across all benchmark systems. For example, noticeable variations exist between the TORA and Unicycle cases, yet the paper does not discuss the reasons behind these differences or identify the conditions under which FABRIC performs most effectively. Moreover, the evaluation is limited to relatively low-dimensional benchmarks, and the absence of scaling analysis with respect to network size, state dimension, or horizon length makes it difficult to assess the generalizability and scalability of the proposed framework to more complex systems.

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